

Hardware Specifications:

Component	Model / Manufacturer	Key Specifications	Interface	Operating Voltage	Qty.	Function in Proposed System	Datasheet
Microcontroller (Transmitter)	ESP32 DevKit V1 (ESP-WROOM-32, Espressif Systems)	Dual-core Xtensa LX6, 240 MHz CPU, 520 kB SRAM, 4 MB Flash, Wi-Fi/BLE	I ² C, SPI, UART, GPIO	3.0–3.6 V (3.3 V nominal)	1	Acquires sensor data, performs CCSDS-inspired packetization, CRC-16 computation, and controls LoRa transmission	ESP32-WROOM-32 Datasheet (Espressif)
Microcontroller (Receiver)	ESP32 DevKit V1 (ESP-WROOM-32, Espressif Systems)	Dual-core Xtensa LX6, 240 MHz CPU, 520 kB SRAM, 4 MB Flash	SPI, UART	3.0–3.6 V	1	Receives LoRa packets, verifies CRC, decodes telemetry, and transfers data to MATLAB	ESP32-WROOM-32 Datasheet (Espressif)
LoRa Transceiver	SX1278 RA-02 (Semtech / Ai-Thinker)	433 MHz ISM band, +20 dBm Tx power, −148 dBm receiver sensitivity, Chirp Spread Spectrum (CSS)	SPI	1.8–3.7 V	2	Wireless telemetry communication between transmitter and receiver	SX1278 Datasheet (Semtech)
Temperature Sensor	TMP102 (Texas Instruments)	Digital temperature sensor, −40°C to +125°C, 0.0625°C resolution, ±0.5°C accuracy (typ.)	I ² C	1.4–3.6 V	1	Measures onboard temperature telemetry	TMP102 Datasheet (Texas Instruments)
Pressure Sensor	BMP280 (Bosch Sensortec)	Pressure: 300–1100 hPa, Temperature: −40°C to +85°C, 24-bit ADC	I ² C	1.71–3.6 V	1	Measures atmospheric pressure	BMP280 Datasheet (Bosch Sensortec)

3-Axis Accelerometer	ADXL345 (Analog Devices)	$\pm 2/\pm 4/\pm 8/\pm 16$ g selectable range, 13-bit resolution, digital output	I ² C	2.0–3.6 V	1	Measures three-axis acceleration and orientation	ADXL345 Datasheet (Analog Devices)
Transmit Antenna	433 MHz Rubber Duck Antenna	Quarter-wave monopole, omnidirectional, nominal gain ≈ 2 dBi, SMA connector	RF	Passive	1	LoRa packet transmission	Manufacturer Specification Sheet (Robu.in)
Receive Antenna	433 MHz Rubber Duck Antenna	Quarter-wave monopole, omnidirectional, nominal gain ≈ 2 dBi, SMA connector	RF	Passive	1	LoRa packet reception	Manufacturer Specification Sheet (Robu.in)